Supplementary Material for "A Reproducible ROI-Robust Evaluation and Mobile Distillation Framework for Ultrasound Lesion Classification"

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1. Supplementary Tables

Table S1 records the validation-performance audit used to reduce test-set model-selection risk. The selected model is retained for the original dataset-specific benchmark, while the main manuscript also reports the stronger MUDD+DA-noMCA robustness variant evaluated on the frozen analysis-label snapshot.

Table S1: Validation-performance audit for the TransXNet-family selection. Values are averages over three TN5000 seeds or five BUSI/AUL training folds.

Variant	TN5000 val AUC	BUSI val AUC	AUL val AUC	Mean val AUC	Selected
TransXNet	0.9577	0.7796	0.7464	0.8279	no
TransXNet-GG	0.9632	0.8011	0.7390	0.8345	no
GGG-noMCA	**0.9700**	0.8957	0.8798	0.9152	no
UL-TransXNet	0.9638	**0.9199**	**0.8827**	**0.9222**	yes

The comparison pool uses the same ROI construction, augmentation family, validation-based thresholding, and reporting scripts, but the unusually low ResNet50 TN5000 result remains a baseline-sanity risk. We retain it for transparency rather than using it as the primary comparison point. The main manuscript therefore compares UL-TransXNet with the strongest observed comparison baseline on each dataset, while the release artifacts include the repeated-run logs and prediction files needed to inspect whether ResNet50 instability comes from this ROI protocol, optimization sensitivity, or implementation details.

Table S2: Complexity–performance trade-off of the comparison pool. CUDA latency is local workstation model-only batch-1 latency. Mean AUC and Mean BalAcc are unweighted averages over TN5000, BUSI, and AUL.

Method	Input	Params	MACs	CUDA ms	TN5000	BUSI	AUL	Mean AUC	Mean BalAcc
MobileNetV3-Large	256	4.86M	0.28G	3.17	0.8282	0.7947	0.7357	0.7862	0.7183
EfficientNet-B0	256	4.66M	0.50G	6.53	0.7856	0.7659	0.6099	0.7205	0.6682
EfficientFormer-L1	224	11.62M	1.31G	3.41	0.9249	0.7917	0.7196	0.8121	0.7330
RepViT-M1.1	256	8.04M	1.80G	5.19	0.8403	0.8028	0.6797	0.7742	0.7020
UL-TransXNet	256	14.37M	2.36G	30.23	**0.9532**	**0.8485**	0.8248	**0.8755**	**0.8037**
DenseNet121	256	7.48M	3.70G	7.19	0.9360	0.8154	0.6729	0.8081	0.7401
ResNet50	256	24.56M	5.40G	3.25	0.6531	0.7519	0.5481	0.6510	0.6190
ConvNeXt-Tiny	256	28.21M	5.82G	3.30	0.9508	0.8050	0.8274	0.8611	0.7651
Swin-T	256	27.91M	7.11G	6.32	0.9497	0.8190	**0.8319**	0.8669	0.7901

Table S3: Two-device Android repeat for the exported teacher and distilled students. Both devices use the paper-log TN5000 analysis-label snapshot, file-mode ROI loading, 30% expanded non-square crops, and five hot runs per model.

Device	Model	Android AUC	Hot total ms	Hot inference ms	P95 total ms
Xiaomi 24129PN74C	EfficientFormer-L1+KD	0.9634	32.690	31.298	37.667
Xiaomi 24129PN74C	EfficientFormer-L1+ECA+KD	0.9576	34.853	33.390	41.103
Xiaomi 24129PN74C	GGG-MCA teacher	0.9433	130.551	128.831	132.464
Samsung SM-X800	EfficientFormer-L1+KD	0.9634	55.954	54.697	67.714
Samsung SM-X800	EfficientFormer-L1+ECA+KD	0.9576	56.159	54.852	58.328
Samsung SM-X800	GGG-MCA teacher	0.9433	168.450	166.806	170.529

Table S4: Offline paired statistical comparison between UL-TransXNet and the strongest comparison baseline in the main benchmark table after recomputation with the frozen paper-log labels. These are repeated-run stability checks, not large-sample clinical significance tests.

Dataset	Baseline	Metric	n	Δ	Run-delta range	p
TN5000	ConvNeXt-Tiny	AUC	3	+0.0024	[-0.0024, +0.0051]	1.0000
TN5000	ConvNeXt-Tiny	BalAcc	3	+0.0058	[-0.0023, +0.0215]	1.0000
BUSI	Swin-T	AUC	5	+0.0294	[+0.0179, +0.0422]	0.0625
BUSI	Swin-T	BalAcc	5	+0.0714	[+0.0460, +0.0989]	0.0625
AUL	Swin-T	AUC	5	-0.0071	[-0.0189, +0.0203]	0.3750
AUL	Swin-T	BalAcc	5	-0.0376	[-0.1117, +0.0869]	0.3750

Table S5: Android baseline context on the Xiaomi 24129PN74C device. AUC values use the paper-log TN5000 analysis-label snapshot; latency is the measured hot end-to-end runtime.

Model	Android AUC	Hot total latency (ms)	Model size (MB)
MobileNetV3-Large	0.8146	15.745	18.528
EfficientFormer-L1	0.9168	34.451	44.303
RepViT-M1.1	0.9035	46.188	30.749
ConvNeXt-Tiny	0.9505	169.662	107.718
GGG-MCA teacher	0.9433	130.551	53.458
EfficientFormer-L1+KD	**0.9634**	**32.690**	44.303
EfficientFormer-L1+ECA+KD	0.9576	34.853	44.304

Table S6: Closed-loop automatic ROI classification summary. Oracle uses annotation boxes, auto uses detector-derived boxes, and full resizes the whole image.

Dataset	Input	AUC	BalAcc	F1	Acc
BUSI	oracle	0.8485	0.7851	0.7695	0.7984
BUSI	auto	0.8393	0.7809	0.7667	0.7969
BUSI	full	0.7575	0.7064	0.6710	0.6961
AUL	oracle	0.8248	0.7407	0.7215	0.7323
AUL	auto	0.8094	0.7358	0.7124	0.7244
AUL	full	0.7104	0.6268	0.6307	0.6866

Table S7: Detector localization quality used for automatic ROI construction. TN5000 values are mean $\pm$ standard deviation across three detector seeds on the official test split; BUSI/AUL values are mean $\pm$ standard deviation across five detector folds on the fixed held-out test sets.

Dataset	Mean IoU	R@0.50	R@0.75	No det.
TN5000	0.7855 ± 0.0033	0.9350 ± 0.0035	0.7880 ± 0.0035	0.0000
BUSI	0.7365 ± 0.0241	0.8403 ± 0.0141	0.6822 ± 0.0548	0.0000
AUL	0.5755 ± 0.0221	0.6740 ± 0.0419	0.4331 ± 0.0201	0.0000

Table S8: Descriptive BUSI duplicate and near-duplicate audit. The audit uses the diagnostic labels from the BUSI label manifest, not the one-class detector labels stored in VOC XML files. It does not alter the public-benchmark split or main reported metrics. Near duplicates require both dHash and aHash Hamming distances at or below the stated threshold.

Hash screen	Exact duplicate pairs	Near-duplicate pairs	Cross-split pairs	Label-conflict pairs	Output directory
Strict (≤ 2)	1	88	26	6	`busi_duplicate_audit_20260510_manifestlabels_t2`
Relaxed (≤ 5)	1	141	41	9	`busi_duplicate_audit_20260510_manifestlabels_t5`

Table S9: Case-level uncertainty and calibration summary for seed/fold-averaged UL-TransXNet predictions recomputed with the frozen paper-log labels. ECE uses 10 confidence bins.

Dataset	AUC 95% CI	BalAcc 95% CI	ECE	Brier
TN5000	0.960 [0.946, 0.973]	0.909 [0.892, 0.930]	0.0226	0.0662
BUSI	0.850 [0.756, 0.932]	0.800 [0.746, 0.884]	0.0538	0.1770
AUL	0.826 [0.733, 0.903]	0.832 [0.760, 0.902]	0.1474	0.1979

Table S10: Case-level diagnostic statistics for seed/fold-averaged UL-TransXNet predictions on the fixed test sets. Brackets are case-bootstrap 95% confidence intervals.

Dataset	AUC	Sens.	Spec.	PPV	NPV	ECE/Brier
TN5000	0.960 [0.946, 0.973]	0.893	0.925	0.968	0.772	0.023/0.066
BUSI	0.850 [0.756, 0.932]	0.711	0.890	0.730	0.880	0.054/0.177
AUL	0.826 [0.733, 0.903]	0.892	0.773	0.881	0.791	0.147/0.198

2. Descriptive Complexity Views

Figure S1 keeps the original accuracy–complexity visualization as a descriptive summary. It is not used as the main evidence for model selection because the vertical axis averages dataset-specific test AUC values from datasets with different organs, sizes, and prevalence. The main manuscript therefore uses protocol-separated tables and mobile latency analysis for the primary claims.

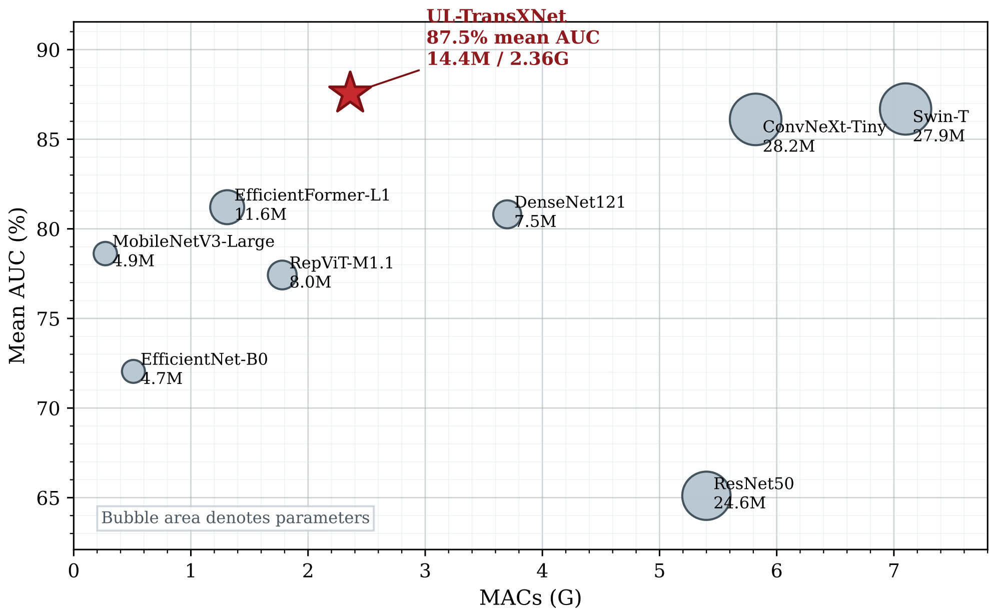

Figure S1: Descriptive mean AUC versus MACs view. Mean AUC is the unweighted average over TN5000, BUSI, and AUL under dataset-specific training; it should not be interpreted as external cross-organ validation.

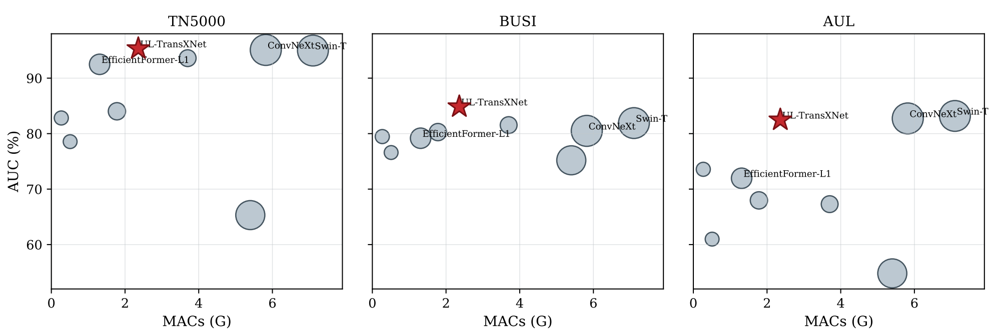

Figure S2: Additional complexity–AUC trade-off visualization from the original benchmark package. This figure is retained for reference but secondary to the protocol-separated result tables.

3. Additional Diagnostic Figures

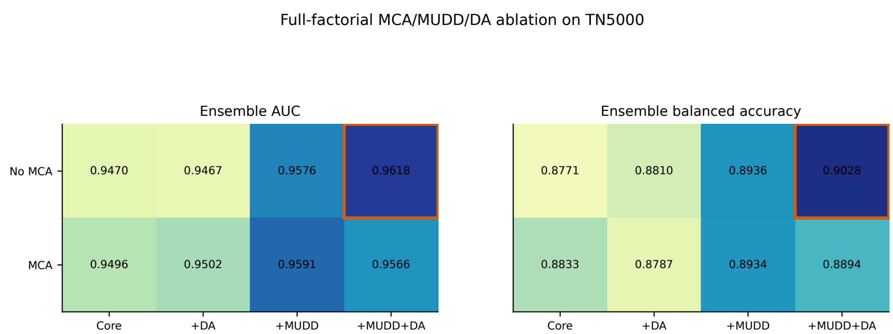

Figure S3: Heatmap view of the full-factorial TN5000 MCA/MUDD/differential-attention ablation. The main manuscript reports the corresponding numerical table.

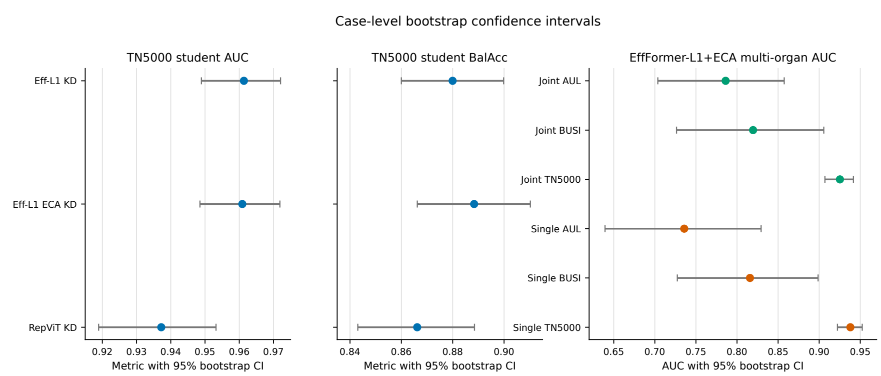

Figure S4: Bootstrap confidence-interval forest plot for selected frozen-snapshot experiments. Case-level intervals are provided as statistical transparency rather than clinical significance claims.

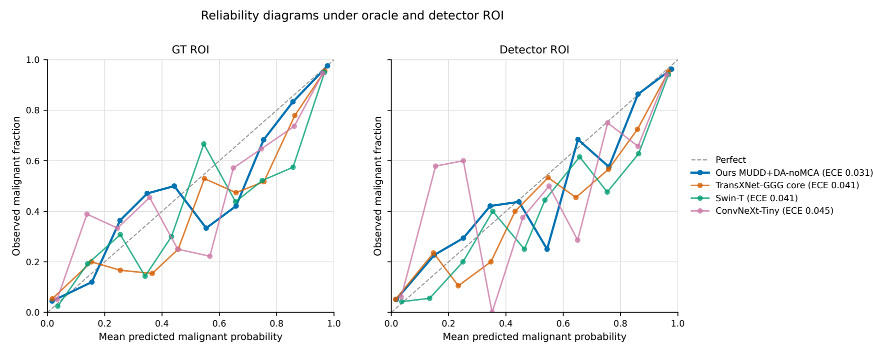

Figure S5: Reliability and calibration diagnostics for oracle and detector-derived ROI settings. These plots support the manuscript's cautious discussion of probability calibration.

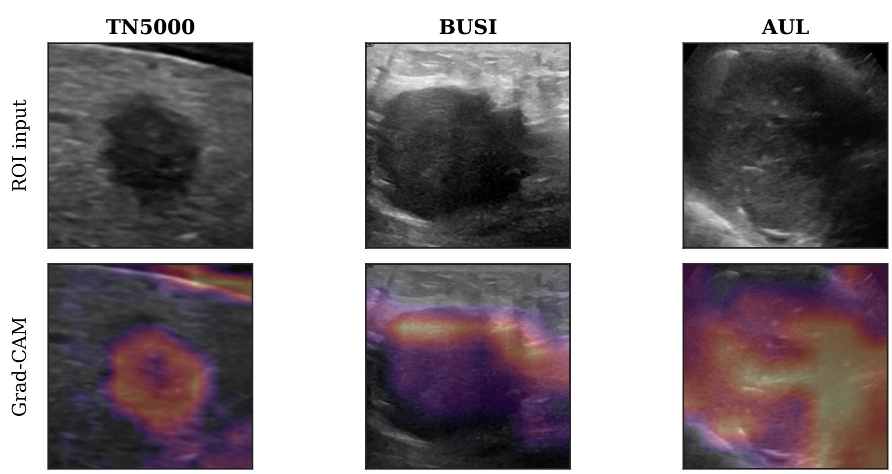

Figure S6: Grad-CAM examples used as qualitative sanity checks. They are not interpreted as causal explanations of model decisions.